

12

Dimensionality Reduction

Projects multidimensional data onto a lower-dimensional space, revealing the data's intrinsic structure

In this chapter, we start from the most basic data matrix and gradually build a deep understanding of principal component analysis. This chapter features a special Gram matrix — the covariance matrix — playing the central role. We'll use the eigenvalue decomposition introduced in the previous chapter to explore the covariance matrix, thereby understanding the structure of the data and achieving dimensionality reduction, compression, and feature extraction.

12.1 Data Matrix



Core skills you will master in this section:

- ▶ In a data matrix, rows represent samples and columns represent features.
- ▶ Use matrix multiplication to compute the mean of each feature column and the overall data centroid.
- ▶ Demeaning moves the data's centroid to the origin, making subsequent covariance analysis more convenient.
- ▶ The covariance matrix is a Gram matrix of centered data, reflecting inner-product structure.
- ▶ In the covariance matrix, the diagonal holds the variances and the off-diagonal holds the covariances.
- ▶ Standardization: subtract the mean and divide by the standard deviation for each column, making the data unitless with mean 0 and standard deviation 1.
- ▶ The linear correlation coefficient normalizes the covariance so that it is unaffected by units, making it easier to compare features.
- ▶ The linear correlation coefficient matrix is the covariance matrix of standardized data, and is also a Gram matrix.

This section starts from the data matrix and walks step by step through the common operations that surround it — such as computing the centroid, centering, computing the covariance matrix, standardization, and computing the linear correlation coefficient matrix. A whole toolbox of linear algebra techniques will come into play!

It's worth noting that both the covariance matrix and the linear correlation coefficient matrix can be viewed as two special Gram matrices of the data matrix.

Data Matrix

A data matrix (data matrix) is a way of organizing all kinds of real-world data into a "row $\times$ column" format.

The most intuitive example is a spreadsheet, where each row represents a sample (sample) and each column represents a feature.

For example, in a table of student grades, each row is a student, and each column might be the score in a subject such as math, Chinese, or English. This is a typical data matrix.

The "Iris" book series generally uses the matrix $\mathbf{X}$ (uppercase, italic, bold) to represent the data matrix.

As shown in Figure 1 (a), the data matrix $\mathbf{X}$ has n rows and D columns.

As shown in Figure 1 (b), the data matrix $\mathbf{X}$'s every column represents the sample data for a feature (or random variable); $\mathbf{X}$ has D features, corresponding to D column vectors. Simply put, a random variable is a variable used to describe the outcome of a random phenomenon.

As shown in Figure 1 (c), the data matrix $\mathbf{X}$'s every row represents a sample; $\mathbf{X}$ has n rows, meaning n samples.

 Please preview (or review) how to compute the sample mean, variance, standard deviation, covariance, and linear correlation coefficient.

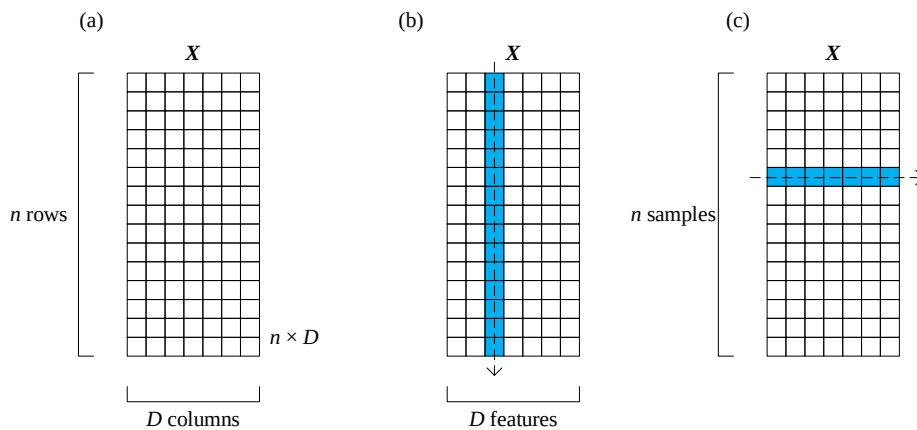


Figure 1. Data Matrix

 Note that in this book, algebraic (scalar) variables are written in lowercase italic, such as x , x_1 , x_2 , y , y_1 , y_2 , and so on; the data matrix is written in uppercase, italic, bold $\mathbf{X}$. Data column vectors are written in lowercase, italic, bold, with a numeric subscript, such as $\mathbf{x}$, $\mathbf{x}_1$, $\mathbf{x}_2$, $\mathbf{y}$, $\mathbf{y}_1$, $\mathbf{y}_2$, and so on; data row vectors (samples) are written in lowercase, italic, bold, with a numeric superscript, such as $\mathbf{x}^{(1)}$, $\mathbf{x}^{(2)}$, $\mathbf{y}^{(1)}$, $\mathbf{y}^{(2)}$, and so on; random variables are written in uppercase, italic, with a numeric subscript, such as X , X_1 , X_2 , Y , Y_1 , Y_2 , and so on.

Row Vectors, Column Vectors

Figure 2 shows data for the first 4 columns (sepal length, sepal width, petal length, petal width) of features from the iris dataset.

Each row of the data (a row vector) represents one iris sample; for example, the row vector of the first row is

$$\mathbf{x}^{(1)} = [5.1 \quad 3.5 \quad 1.4 \quad 0.2]$$

 Note that all the data in Figure 2 is in centimeters (cm); Figure 2 does not include the iris label column.

Figure 2 — each column vector represents a feature, such as sepal length, sepal width, petal length, or petal width.

For example, the first column is the sepal length data

$$\mathbf{x}_1 = \begin{bmatrix} 5.1 \\ 4.9 \\ 4.7 \\ \vdots \end{bmatrix}$$

This column has 150 values, corresponding to 150 iris samples.

Based on this data, various statistical computations can be performed, such as computing the mean, variance, and standard deviation. And based on two sets of column-vector data, we can compute the covariance and the linear correlation coefficient. These are all topics introduced later in this section.

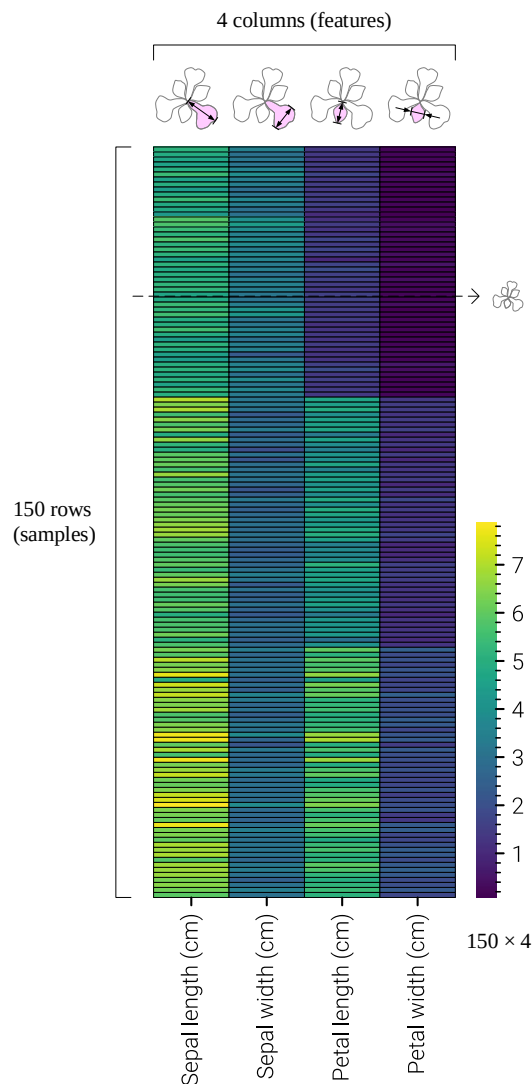


Figure 2. Heatmap of the Iris Data Matrix, in Centimeters, Excluding the Final Label Column

Chapter 10, Section 1 of this book uses a planar scatter plot to show two features' data, and Chapter 10, Section 3 uses a 3D scatter plot to show three features' data. For the 4 features' data shown in Figure 2, in order to show how the data is distributed, we need a pairwise scatter plot.

Figure 3 shows the pairwise scatter plot drawn from the iris data.

Figure 3 is also equivalent to a 4×4 matrix, where each element of the matrix is a subplot.

Figure 3 — the off-diagonal subplots are planar scatter plots, which you should already be quite familiar with.

Figure 3 — the diagonal subplots are kernel density estimation plots (Kernel Density Estimation plot, KDE plot); this type of plot is a visualization tool that approximates the data's distribution with a smooth curve, showing the data's probability density distribution. It is smoother than a histogram and helps in understanding the data's central tendency and distribution shape.

We can imagine the data in Figure 2 as points scattered in 4D space; projecting them onto different planes gives the subplots in Figure 3.

For example, projecting the data matrix $\mathbf{X}$ onto the x_1 axis is equivalent to extracting $\mathbf{X}$'s first column $\mathbf{x}_1$, corresponding to the matrix multiplication

$$\mathbf{X} @ \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \mathbf{x}_3 \quad \cdots \quad \mathbf{x}_D] @ \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \mathbf{x}_1$$

This data is used to draw the image in Figure 2 at row 1, column 1.

The data matrix $\mathbf{X}$ projected onto the x_1x_2 plane is equivalent to extracting $\mathbf{X}$'s first two columns

$$\mathbf{X} @ \begin{bmatrix} 1 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \mathbf{x}_3 \quad \cdots \quad \mathbf{x}_D] @ \begin{bmatrix} 1 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = [\mathbf{x}_1 \quad \mathbf{x}_2]$$

This data is used to draw the image in Figure 2 at row 2, column 1; transposing this image gives the image in Figure 2 at row 1, column 2.

We're sure you've already noticed that Figure 3 is equivalent to a symmetric matrix — we really only need the subplots on the main diagonal, plus the six scatter plots either above or below the main diagonal.

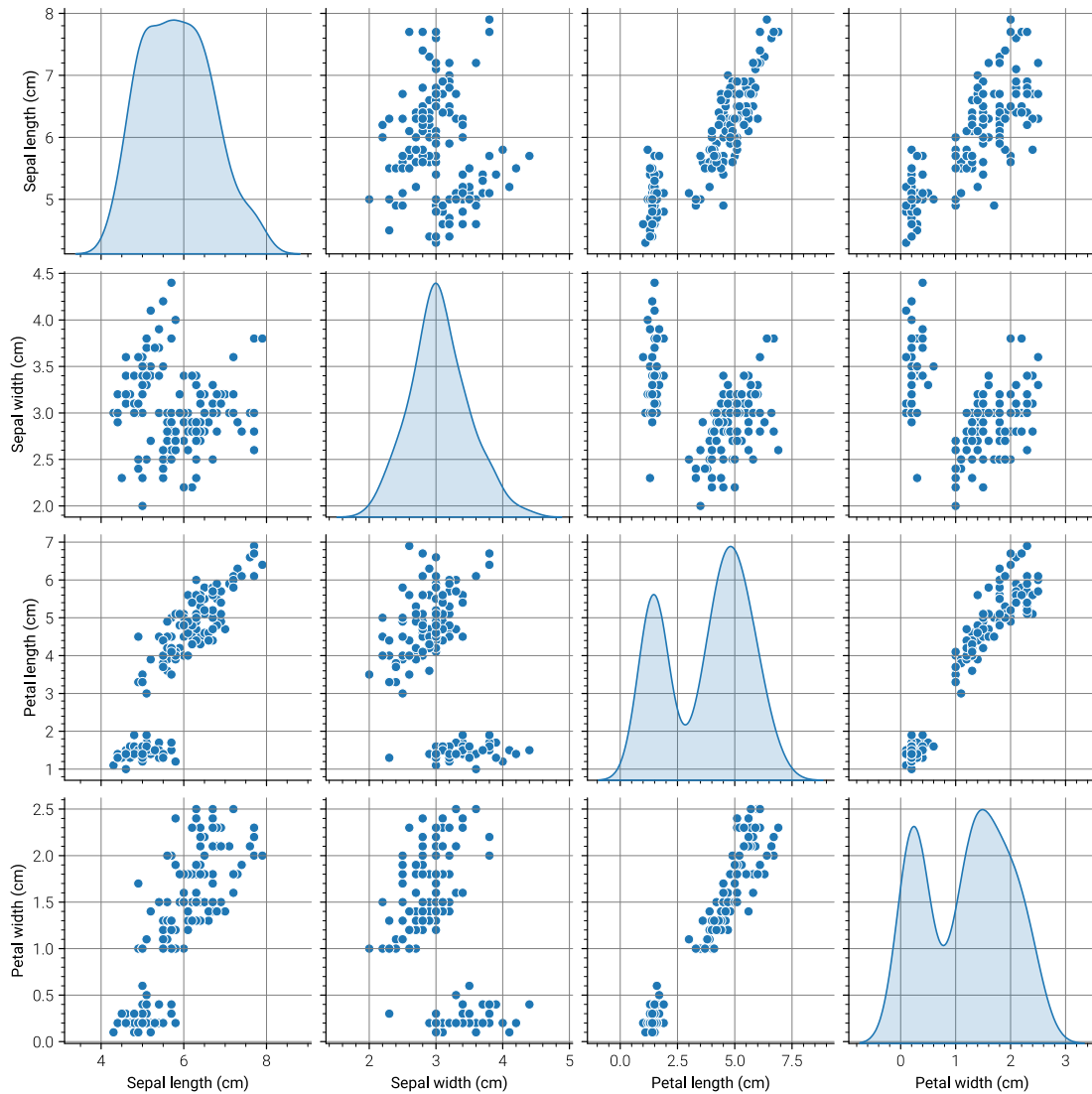


Figure 3. Pairwise Scatter Plot of the Iris Data

Mean, Centroid

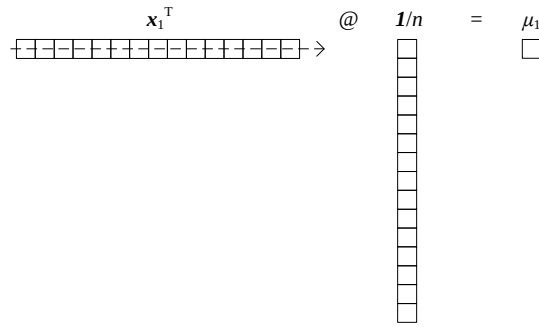
In data analysis, the mean (mean, average) is the most common measure of central tendency; it represents the "average" position of a set of data.

For the iris dataset, $\mathbf{x}_1$'s mean μ_1 is the sum of all the sepal-length data-point values divided by the number of data points ($n = 150$), corresponding to the matrix multiplication

$$\mu_1 = \frac{\mathbf{1}^T \mathbf{x}_1}{n} = \frac{\mathbf{x}_1^T \mathbf{1}}{n} = \frac{\mathbf{1} \cdot \mathbf{x}_1}{n} = \frac{\mathbf{x}_1 \cdot \mathbf{1}}{n} = \frac{\sum_{i=1}^n x_{i,1}}{n}$$

⚠ Note that in the equation above, the all-ones column vector $\mathbf{1}$ has n rows.

❓ Please use this section's companion code to compute the mean of each feature.

Figure 4. Computing $\mathbf{x}_1$'s Mean

It's easy to compute the sample means of the iris dataset's four features

$$\left\{ \begin{array}{l} \mu_1 = 5.843 \\ \text{Sepal length} \\ \mu_2 = 3.057 \\ \text{Sepal width} \\ \mu_3 = 3.758 \\ \text{Petal length} \\ \mu_4 = 1.199 \\ \text{Petal width} \end{array} \right.$$

⚠ Note that in the equation above, in this book, the centroid $\boldsymbol{\mu}$ is a column vector; the mean's units are all centimeters, consistent with the sample data's units.

If the equation above is not divided by n , the result is the sum of the sample values, that is

$$\sum_{i=1}^n x_{i,1} = \mathbf{1}^T \mathbf{x}_1 = \mathbf{x}_1^T \mathbf{1} = \mathbf{1} \cdot \mathbf{x}_1 = \mathbf{x}_1 \cdot \mathbf{1} = n\mu_1$$

The centroid (centroid), $\boldsymbol{\mu}$, is a measure of location for multivariate data – it is the geometric center of all the sample data points, that is

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_D \end{bmatrix}$$

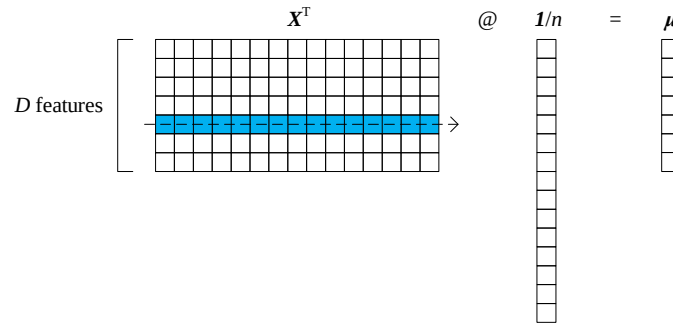
Geometrically, the centroid is the point that "balances" all the sample points.

⚠ Note that in this book, the centroid $\boldsymbol{\mu}$ is a column vector.

As shown in Figure 5, the data matrix $\mathbf{X}$'s centroid can be obtained via the following matrix multiplication

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_D \end{bmatrix} = \frac{\mathbf{X}^T \mathbf{1}}{n}$$

⚠ Note that in the equation above, the all-ones column vector $\mathbf{1}$ has n rows.

Figure 5. Computing the Data Matrix X 's Centroid

The centroid of the iris data is

$$\boldsymbol{\mu} = \begin{bmatrix} 5.843 \\ 3.057 \\ 3.758 \\ 1.199 \end{bmatrix} \quad (5)$$

Figure 6 – we've marked the centroid's location with a red $\times$ on the pairwise scatter plot.

Projecting the centroid $\boldsymbol{\mu}$ onto the x_1 axis gives μ_1

$$\mu_1 = \boldsymbol{\mu}^T \mathbf{e}_1 = \mathbf{e}_1^T \boldsymbol{\mu} = \boldsymbol{\mu} = [1 \quad 0 \quad 0 \quad 0] @ \begin{bmatrix} 5.843 \\ 3.057 \\ 3.758 \\ 1.199 \end{bmatrix} = 5.843$$

? Please think about how to use the centroid to compute the means of the other features.

Projecting the centroid $\boldsymbol{\mu}$ onto the x_1x_2 plane gives $[\mu_1, \mu_2]^T$, corresponding to the following matrix multiplication

$$\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} @ \begin{bmatrix} 5.843 \\ 3.057 \\ 3.758 \\ 1.199 \end{bmatrix} = \begin{bmatrix} 5.843 \\ 3.057 \end{bmatrix}$$

This centroid projection corresponds to the $\times$ position in Figure 6's row 2, column 1 subplot.

? Please think about how to compute the centroid projection positions for the other subplots in Figure 6.

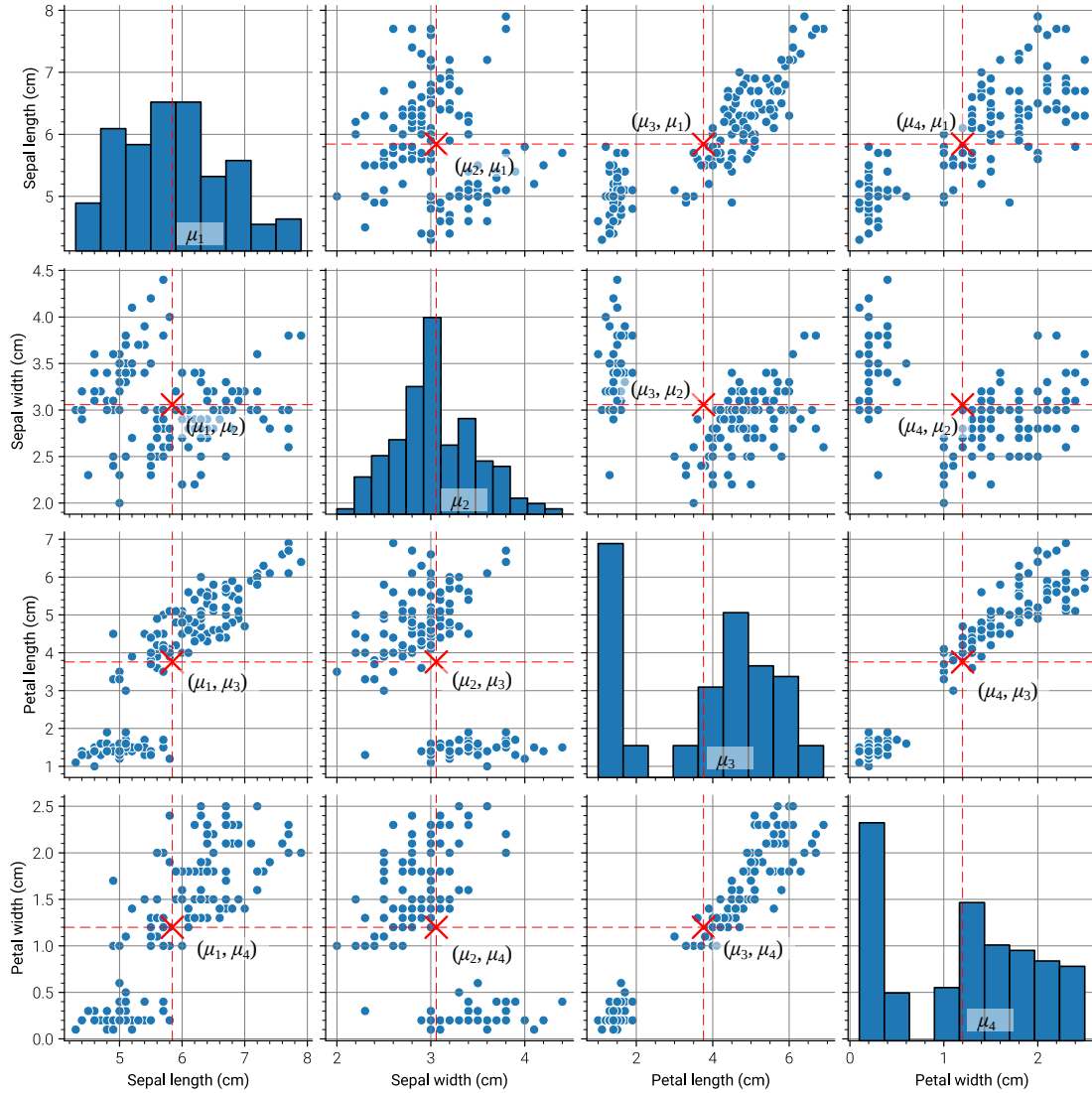


Figure 6. Pairwise Scatter Plot of the Iris Data, with the Red × Marking the Centroid

Data Centering

Data centering (centralize), also called demeaning (demean), refers to subtracting the column mean from every value in each column of the data matrix $\mathbf{X}$, making the data's mean become zero.

This process of (column) centering the data does not change the shape of the data's distribution, but shifts the data so that it is centered on the origin.

Geometrically, this means moving the data's centroid $\boldsymbol{\mu}$ to the origin $\mathbf{0}$.

For example, in the data matrix $\mathbf{X}$'s first column, subtracting its mean μ_1 corresponds to the matrix multiplication

$$\mathbf{x}_1 - \mu_1 \mathbf{1} = \mathbf{x}_1 - \underbrace{\frac{1}{n} \mathbf{1} \mathbf{1}^T}_{\mathbf{M}} \mathbf{x}_1 = \left(\mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) \mathbf{x}_1$$

Note that in the equation above, the all-ones column vector $\mathbf{1}$ has n rows.

Let

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Comments and corrections are welcome. The book's dedicated email address: jiang.visualize.ml@gmail.com

$$\mathbf{M} = \mathbf{I} - \frac{1}{n} \mathbf{1}\mathbf{1}^T$$

We call $\mathbf{M}$ the centering matrix, or the demeaning matrix. The matrix $\mathbf{M}$ is idempotent, meaning it satisfies $\mathbf{M}\mathbf{M} = \mathbf{M}$.

The data matrix $\mathbf{X}$'s column centering can be computed via the equation below

$$\mathbf{X}_c = \mathbf{X} - \frac{1}{n} \mathbf{1}\mathbf{1}^T \mathbf{X} = \underbrace{\left(\mathbf{I} - \frac{1}{n} \mathbf{1}\mathbf{1}^T \right)}_{\mathbf{M}} \mathbf{X} = \mathbf{M}\mathbf{X}$$

Geometrically, the equation above amounts to a translation. Building on the "translation" above,

The data matrix $\mathbf{X}$'s row centering can be computed this way

$$\mathbf{X} \underbrace{\left(\mathbf{I} - \frac{1}{n} \mathbf{1}\mathbf{1}^T \right)}_{\mathbf{M}} = \mathbf{X}\mathbf{M}$$

⚠ Note that in the equation above and the one before it, the shapes of $\mathbf{M}$ are different.

Using the broadcasting rule (broadcasting), the equation above can be written as

$$\mathbf{X}_c = \mathbf{X} - \boldsymbol{\mu}^T$$

The equation above amounts to performing the subtraction for every row vector.

The broadcasting rule in matrix operations refers to automatically expanding the smaller-dimension matrix when dimensions don't match, so that the shapes align and element-wise operations can be carried out.

Figure 7 shows the heatmap of the centered iris data matrix.

Figure 8 shows the pairwise scatter plot of the centered data. In the planar scatter subplots, we find that all centroid projections lie at the origin — that is, the centroid projection is the zero vector $\mathbf{0}$.

Comparing the diagonal subplots of Figure 6 and Figure 8, we find that the data's distribution hasn't changed at all — only the mean's position has shifted.

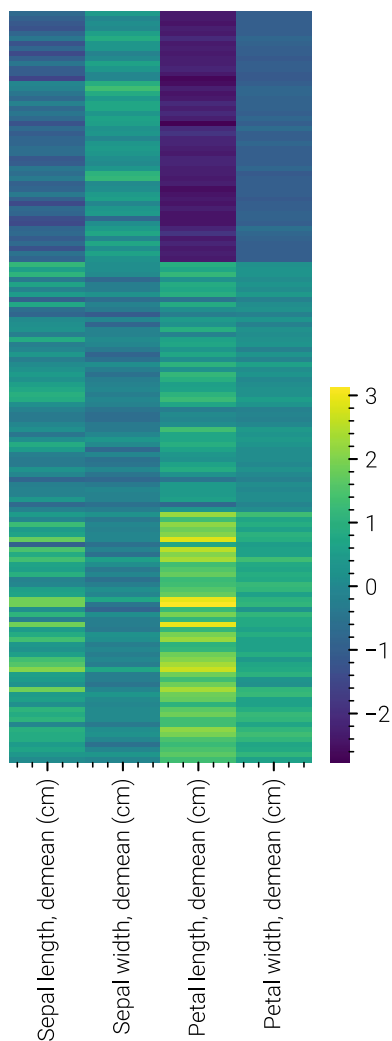


Figure 7. Heatmap of the Iris Data (Centered, Demeaned) Matrix, in Centimeters

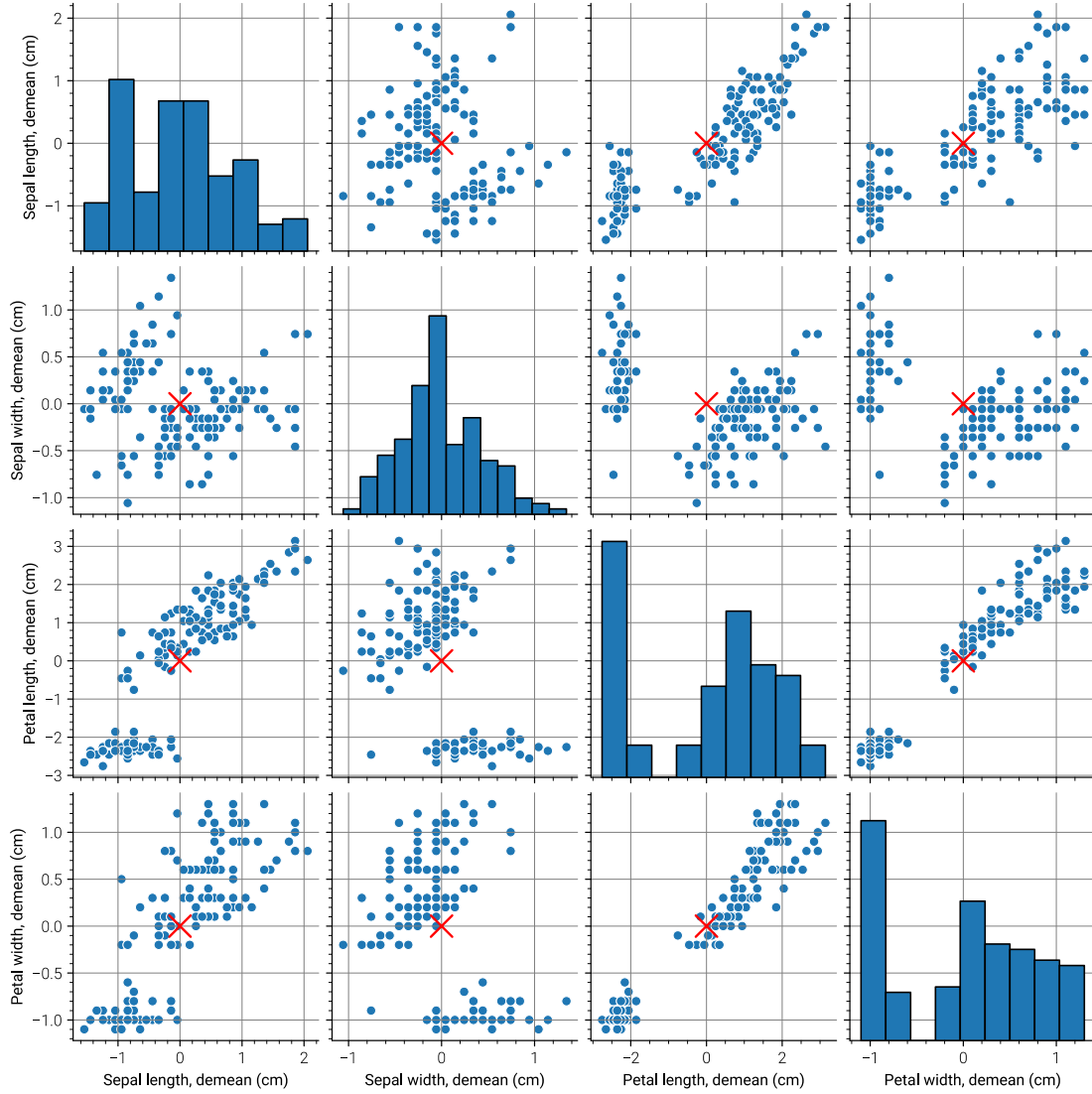


Figure 8. Pairwise Scatter Plot of the Iris Data (Centered, Demeaned), with the Red × Marking the Centroid

Covariance Matrix

As shown in Figure 9, once we have the centered data matrix $\mathbf{X}_c$, we can compute the covariance matrix, corresponding to the matrix multiplication

$$\Sigma = \frac{\mathbf{X}_c^T \mathbf{X}_c}{n-1}$$

Computing $\mathbf{X}_c$'s Gram matrix $\mathbf{X}_c^T \mathbf{X}_c$, and scaling by the factor above $1/(n-1)$ gives the covariance matrix. Additionally, if n is large enough, n can be used in place of $n-1$, with negligible effect.

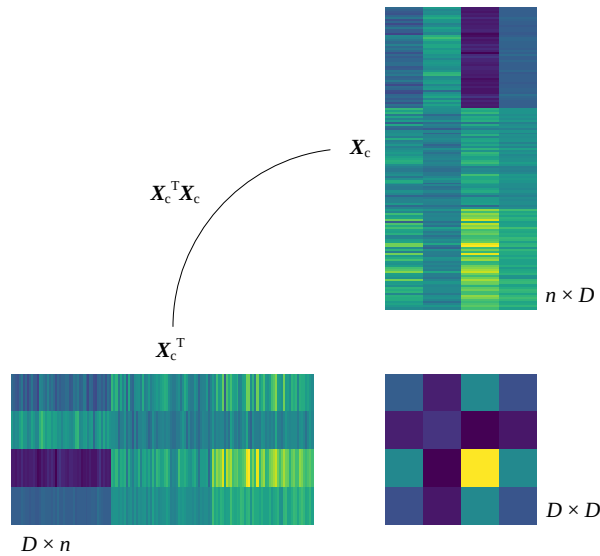
Figure 9. Computing the Sample Covariance Matrix of $\mathbf{X}$ Not Accounting for $n - 1$

Figure 10 shows the specific values of the iris data's covariance matrix.

Note that in the covariance matrix, the units of the variance and covariance are square centimeters.

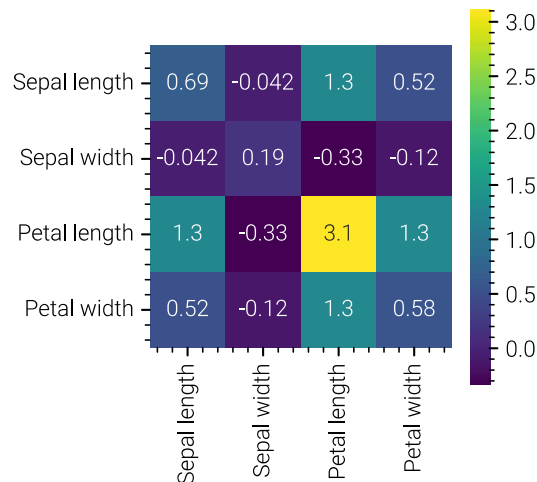


Figure 10. Covariance Matrix, in Square Centimeters

Variance, Covariance, Linear Correlation Coefficient

As shown in Figure 11, the covariance matrix can be viewed as being made up of two parts: variance (variance) and covariance (covariance). Variance consists of the diagonal elements of the covariance matrix, and covariance consists of its off-diagonal elements:

$$\Sigma = \begin{bmatrix} \sigma_{1,1} & \sigma_{1,2} & \cdots & \sigma_{1,D} \\ \sigma_{2,1} & \sigma_{2,2} & \cdots & \sigma_{2,D} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{D,1} & \sigma_{D,2} & \cdots & \sigma_{D,D} \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & \rho_{1,2}\sigma_1\sigma_2 & \cdots & \rho_{1,D}\sigma_1\sigma_D \\ \rho_{1,2}\sigma_1\sigma_2 & \sigma_2^2 & \cdots & \rho_{2,D}\sigma_2\sigma_D \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{1,D}\sigma_1\sigma_D & \rho_{2,D}\sigma_2\sigma_D & \cdots & \sigma_D^2 \end{bmatrix}$$

Variance describes the dispersion of the data along a given feature, while covariance captures the linear correlation between a pair of features. In the equation above, ρ is the linear correlation coefficient (linear correlation coefficient, Pearson correlation coefficient), often abbreviated as the correlation coefficient.

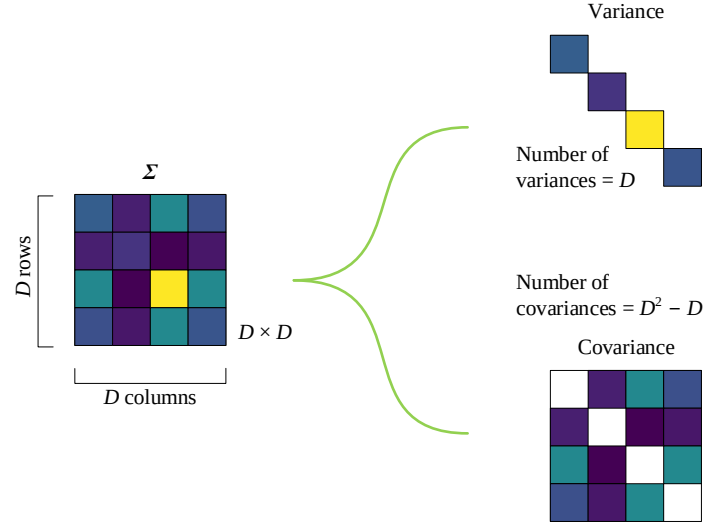


Figure 11. The Covariance Matrix Is Made Up of Variance and Covariance

It is evident that the covariance matrix is symmetric:

$$\Sigma = \Sigma^T$$

Expanding the data matrix $\mathbf{X}$ into a set of column vectors $[\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_D]$, the equation above can be rearranged as:

$$\begin{aligned} \Sigma &= \frac{1}{n-1} \begin{bmatrix} \mathbf{x}_1 - \mu_1 & \mathbf{x}_2 - \mu_2 & \cdots & \mathbf{x}_D - \mu_D \end{bmatrix}^T \begin{bmatrix} \mathbf{x}_1 - \mu_1 & \mathbf{x}_2 - \mu_2 & \cdots & \mathbf{x}_D - \mu_D \end{bmatrix} \\ &= \frac{1}{n-1} \begin{bmatrix} (\mathbf{x}_1 - \mu_1)^T (\mathbf{x}_1 - \mu_1) & (\mathbf{x}_1 - \mu_1)^T (\mathbf{x}_2 - \mu_2) & \cdots & (\mathbf{x}_1 - \mu_1)^T (\mathbf{x}_D - \mu_D) \\ (\mathbf{x}_2 - \mu_2)^T (\mathbf{x}_1 - \mu_1) & (\mathbf{x}_2 - \mu_2)^T (\mathbf{x}_2 - \mu_2) & \cdots & (\mathbf{x}_2 - \mu_2)^T (\mathbf{x}_D - \mu_D) \\ \vdots & \vdots & \ddots & \vdots \\ (\mathbf{x}_D - \mu_D)^T (\mathbf{x}_1 - \mu_1) & (\mathbf{x}_D - \mu_D)^T (\mathbf{x}_2 - \mu_2) & \cdots & (\mathbf{x}_D - \mu_D)^T (\mathbf{x}_D - \mu_D) \end{bmatrix} \end{aligned}$$

Let's look separately at the variance and covariance computations in the equation above.

For example, the equation below computes $\mathbf{x}_1$'s variance

$$\text{var}(\mathbf{x}_1) = \sigma_{1,1} = \sigma_1^2 = \frac{(\mathbf{x}_1 - \mu_1)^T (\mathbf{x}_1 - \mu_1)}{n-1}$$

Have you noticed that the equation above is essentially a vector inner product, that is

$$\text{var}(\mathbf{x}_1) = \sigma_{1,1} = \sigma_1^2 = \frac{(\mathbf{x}_1 - \mu_1) \cdot (\mathbf{x}_1 - \mu_1)}{n-1} = \frac{\langle \mathbf{x}_1 - \mu_1, \mathbf{x}_1 - \mu_1 \rangle}{n-1}$$

Taking the square root of the variance gives the standard deviation (standard deviation), that is

$$\text{std}(\mathbf{x}_1) = \sigma_1 = \sqrt{\text{var}(\mathbf{x}_1)} = \sqrt{\frac{(\mathbf{x}_1 - \mu_1)^T (\mathbf{x}_1 - \mu_1)}{n-1}}$$

Note that the standard deviation's units match those of the original data and the mean. For example, the standard deviations of the iris data's four features are all in centimeters.

The equation below computes the covariance between $\mathbf{x}_1$ and $\mathbf{x}_2$

$$\text{cov}(\mathbf{x}_1, \mathbf{x}_2) = \sigma_{1,2} = \rho_{1,2} \sigma_1 \sigma_2 = \frac{(\mathbf{x}_1 - \mu_1)^T (\mathbf{x}_2 - \mu_2)}{n-1}$$



can also be written as a vector inner product; please work this out on your own.

The linear correlation coefficient can be computed via the equation below

$$\rho_{1,2} = \frac{\text{cov}(\mathbf{x}_1, \mathbf{x}_2)}{\sigma_1 \sigma_2}$$

The linear correlation coefficient ranges from -1 to 1; it is essentially a normalization of the covariance so that it no longer depends on the scale of the variables themselves.

Although covariance can reflect the direction and strength of the linear relationship between two variables, its magnitude is affected by the variables' units and scale, making it hard to compare. The correlation coefficient, by dividing by the two variables' standard deviations, standardizes the covariance into a fixed range, so that regardless of the variables' ranges, the degree of linear correlation between them can be measured on a consistent scale.

As shown in Figure 12, a linear correlation coefficient of 1 indicates a perfect positive correlation, -1 indicates a perfect negative correlation, and a value close to 0 indicates almost no linear relationship.

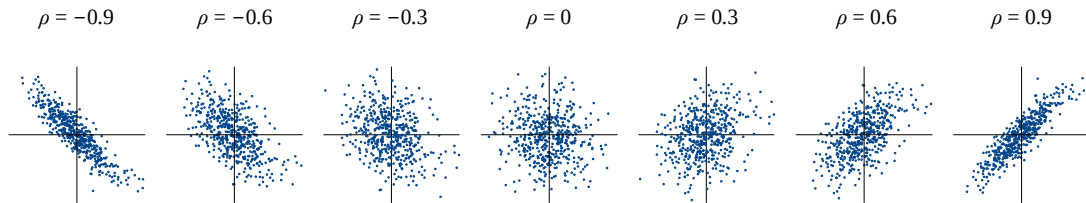


Figure 12. Random-Number Scatter Distribution as the Linear Correlation Coefficient Varies

Centroid at the Origin

Specifically, when all means are 0, that is $[\mu_1, \mu_2, \dots, \mu_D]^T = [0, 0, \dots, 0]^T$, this means the data's centroid is at the origin. Writing $\mathbf{X}$ as a set of column vectors, can be written as:

$$\Sigma = \frac{\mathbf{X}^T \mathbf{X}}{n-1} = \frac{\mathbf{G}}{n-1} = \frac{1}{n-1} \begin{bmatrix} \mathbf{x}_1^T \mathbf{x}_1 & \mathbf{x}_1^T \mathbf{x}_2 & \cdots & \mathbf{x}_1^T \mathbf{x}_D \\ \mathbf{x}_2^T \mathbf{x}_1 & \mathbf{x}_2^T \mathbf{x}_2 & \cdots & \mathbf{x}_2^T \mathbf{x}_D \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{x}_D^T \mathbf{x}_1 & \mathbf{x}_D^T \mathbf{x}_2 & \cdots & \mathbf{x}_D^T \mathbf{x}_D \end{bmatrix}$$

Using vector inner products, the equation above can be written as:

$$\Sigma = \frac{1}{n-1} \begin{bmatrix} \langle \mathbf{x}_1, \mathbf{x}_1 \rangle & \langle \mathbf{x}_1, \mathbf{x}_2 \rangle & \cdots & \langle \mathbf{x}_1, \mathbf{x}_D \rangle \\ \langle \mathbf{x}_2, \mathbf{x}_1 \rangle & \langle \mathbf{x}_2, \mathbf{x}_2 \rangle & \cdots & \langle \mathbf{x}_2, \mathbf{x}_D \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{x}_D, \mathbf{x}_1 \rangle & \langle \mathbf{x}_D, \mathbf{x}_2 \rangle & \cdots & \langle \mathbf{x}_D, \mathbf{x}_D \rangle \end{bmatrix}$$

The equation above is the first perspective on matrix multiplication.

Similarly, when the data's centroid is at the origin, writing $\mathbf{X}$ as a set of row vectors, the equation above can be written as:

$$\begin{aligned} \Sigma &= \frac{\mathbf{X}^T \mathbf{X}}{n-1} = \frac{1}{n-1} \begin{bmatrix} \mathbf{x}^{(1)T} & \mathbf{x}^{(2)T} & \cdots & \mathbf{x}^{(n)T} \end{bmatrix} \begin{bmatrix} \mathbf{x}^{(1)} \\ \mathbf{x}^{(2)} \\ \vdots \\ \mathbf{x}^{(n)} \end{bmatrix} \\ &= \frac{1}{n-1} (\mathbf{x}^{(1)T} \mathbf{x}^{(1)} + \mathbf{x}^{(2)T} \mathbf{x}^{(2)} + \cdots + \mathbf{x}^{(n)T} \mathbf{x}^{(n)}) = \frac{1}{n-1} \sum_{i=1}^n \mathbf{x}^{(i)T} \mathbf{x}^{(i)} \end{aligned}$$

Two Perspectives on Matrix Multiplication

Below, we'll look at the equation above through the two perspectives on matrix multiplication.

According to the first perspective on matrix multiplication, assuming the centroid is the zero vector, writing $\mathbf{X}_c$ as $[\mathbf{x}_1 \ \mathbf{x}_2 \ \cdots \ \mathbf{x}_D]$, the equation above can be expanded as follows.

$$\text{var}(\mathbf{X}) = \Sigma = \frac{1}{n-1} \begin{bmatrix} \mathbf{x}_1^T \mathbf{x}_1 & \mathbf{x}_1^T \mathbf{x}_2 & \cdots & \mathbf{x}_1^T \mathbf{x}_D \\ \mathbf{x}_2^T \mathbf{x}_1 & \mathbf{x}_2^T \mathbf{x}_2 & \cdots & \mathbf{x}_2^T \mathbf{x}_D \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{x}_D^T \mathbf{x}_1 & \mathbf{x}_D^T \mathbf{x}_2 & \cdots & \mathbf{x}_D^T \mathbf{x}_D \end{bmatrix} = \frac{1}{n-1} \begin{bmatrix} \langle \mathbf{x}_1, \mathbf{x}_1 \rangle & \langle \mathbf{x}_1, \mathbf{x}_2 \rangle & \cdots & \langle \mathbf{x}_1, \mathbf{x}_D \rangle \\ \langle \mathbf{x}_2, \mathbf{x}_1 \rangle & \langle \mathbf{x}_2, \mathbf{x}_2 \rangle & \cdots & \langle \mathbf{x}_2, \mathbf{x}_D \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{x}_D, \mathbf{x}_1 \rangle & \langle \mathbf{x}_D, \mathbf{x}_2 \rangle & \cdots & \langle \mathbf{x}_D, \mathbf{x}_D \rangle \end{bmatrix}$$

Note that, for convenience in writing the formula, we assume that in the equation above, $\mathbf{x}_j$ ($j = 1, 2, \dots, D$) has already been centered, i.e., demeaned.

As shown in Figure 13, the covariance matrix's main-diagonal elements are $\mathbf{x}_j^T \mathbf{x}_j$, which is equivalent to the vector inner product $\langle \mathbf{x}_j, \mathbf{x}_j \rangle$, and also equivalent to the vector $\mathbf{x}_j$'s squared L2 norm $\|\mathbf{x}_j\|_2^2$.

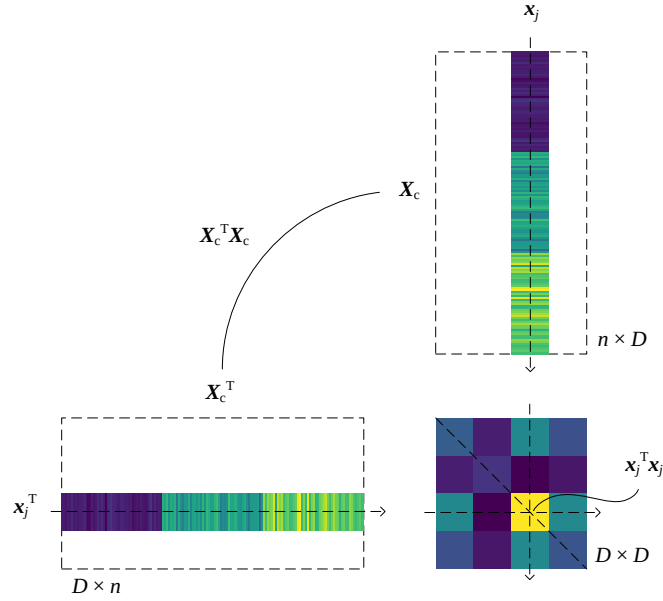


Figure 13. The Covariance Matrix's Main-Diagonal Elements

As shown in Figure 14, the covariance matrix's off-diagonal elements are $\mathbf{x}_j^T \mathbf{x}_k$ ($j \neq k$), which is equivalent to the vector inner product $\langle \mathbf{x}_j, \mathbf{x}_k \rangle$. Clearly, $\mathbf{x}_j^T \mathbf{x}_k = \mathbf{x}_k^T \mathbf{x}_j$, that is $\langle \mathbf{x}_j, \mathbf{x}_k \rangle = \langle \mathbf{x}_k, \mathbf{x}_j \rangle$; this also tells us that the covariance matrix is symmetric.

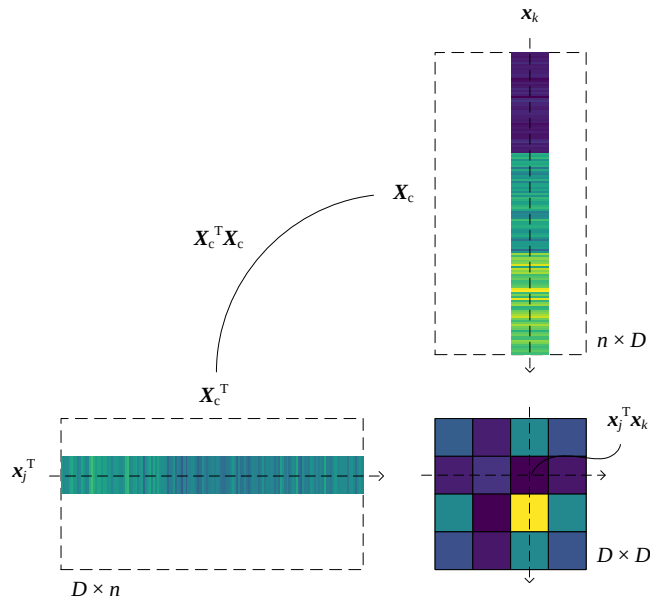


Figure 14. The Covariance Matrix's Off-Diagonal Elements

It's precisely because the covariance matrix is symmetric that, to reduce the amount of information stored, we only need the portion of the matrix shown in Figure 15 – the (variance + covariance) data. Whether it's the lower or upper triangular matrix, we retain D variances and $D(D - 1)/2$ covariances. In other words, we retain $D(D + 1)/2$ elements and eliminate $D(D - 1)/2$ redundant

elements. Using combinatorics, we can easily see that the equation above $C_D^2 = \frac{D(D-1)}{2}$ represents the number of ways to choose any 2 features out of D features.

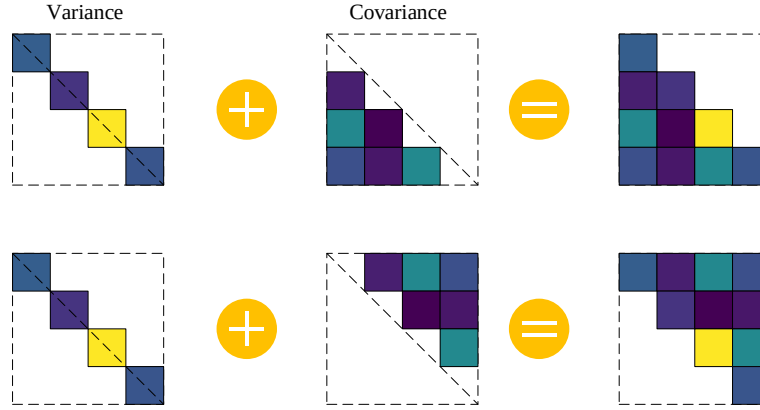


Figure 15. Eliminating Redundant Elements from the Covariance Matrix

For convenience of discussion, again assuming the centroid is the zero vector, according to the

second perspective on matrix multiplication, writing $\mathbf{X}_c$ as $\begin{bmatrix} \mathbf{x}^{(1)} \\ \mathbf{x}^{(2)} \\ \vdots \\ \mathbf{x}^{(n)} \end{bmatrix}$, the equation above can be expanded as the sum of n rank-one matrices, that is

$$\mathbf{\Sigma} = \frac{1}{n-1} \left[\left(\mathbf{x}^{(1)} \right)^T \mathbf{x}^{(1)} + \left(\mathbf{x}^{(2)} \right)^T \mathbf{x}^{(2)} + \dots + \left(\mathbf{x}^{(n)} \right)^T \mathbf{x}^{(n)} \right] = \frac{1}{n-1} \sum_{i=1}^n \left(\mathbf{x}^{(i)} \right)^T \mathbf{x}^{(i)}$$

If the equation above $\mathbf{x}^{(i)}$ is not the zero vector, each $\left(\mathbf{x}^{(i)} \right)^T \mathbf{x}^{(i)}$ is a rank-one matrix with shape $D \times D$.

As shown in Figure 16, the equation above amounts to averaging over the $n \left(\mathbf{x}^{(i)} \right)^T \mathbf{x}^{(i)}$ terms;

moreover, every sample point carries the same weight $\frac{1}{n-1}$.

Although the equation above $\left(\mathbf{x}^{(i)} \right)^T \mathbf{x}^{(i)}$'s rank is 1, the covariance matrix $\mathbf{\Sigma}$'s rank is at most D, that is $\text{rank}(\mathbf{\Sigma}) \leq D$.

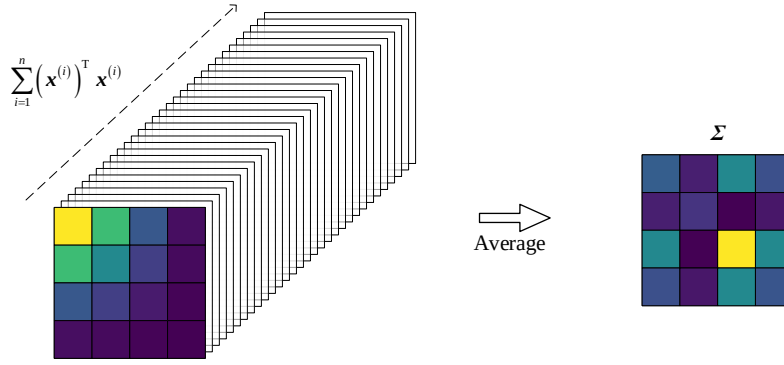


Figure 16. The Covariance Matrix Can Be Viewed as the Average of n Rank-One Matrices

Standardization

Standardizing data (standardization) converts features on different scales onto the same scale, typically by subtracting the mean and dividing by the standard deviation, so that the data has mean 0 and standard deviation 1. This makes it easier to compare different features or to use in algorithms that are sensitive to scale.

Let the diagonal square matrix $\mathbf{D}$ be

$$\mathbf{D} = \text{diag}(\text{diag}(\boldsymbol{\Sigma}))^{\frac{1}{2}} = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_D \end{bmatrix}$$

The inner $\text{diag}()$ extracts the covariance matrix's main-diagonal elements, i.e., extracts the variances, giving a 1D array; the outer $\text{diag}()$ constructs a diagonal square matrix from that 1D array.

The diagonal square matrix's main-diagonal elements are the standard deviations of each feature. With the linear algebra foundation from earlier in this book, you should already understand that this diagonal square matrix's role amounts to scaling.

Building on the "translation" above, scaling further gives the standardized data matrix $\mathbf{Z}$, that is

$$\mathbf{Z} = \mathbf{X}_c \mathbf{D}^{-1} = \underbrace{\left(\mathbf{I} - \frac{1}{n} \mathbf{1}\mathbf{1}^T \right)}_{\mathbf{M}} \mathbf{X} \mathbf{D}^{-1} = \mathbf{M} \mathbf{X} \mathbf{D}^{-1}$$

where

$$\mathbf{D}^{-1} = \text{diag}(\text{diag}(\boldsymbol{\Sigma}))^{-\frac{1}{2}} = \begin{bmatrix} 1/\sigma_1 & 0 & \cdots & 0 \\ 0 & 1/\sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1/\sigma_D \end{bmatrix}$$

Geometrically, the equation above amounts to "translation $\rightarrow$ scaling".

Figure 17 shows the pairwise scatter plot of the standardized iris data.

The standardized data is converted into a unitless (unitless) form because the standardization operation subtracts each value's feature mean and then divides by the standard deviation.

The result after this processing represents "how many standard deviations away from the mean", rather than the original physical quantity, so it no longer carries the original units.

For example, after standardizing sepal length from centimeters, the result represents how many standard deviations that iris sample's sepal length is above or below the average length, rather than "how many centimeters". This allows data with different units and different dimensions to be compared and modeled on the same scale

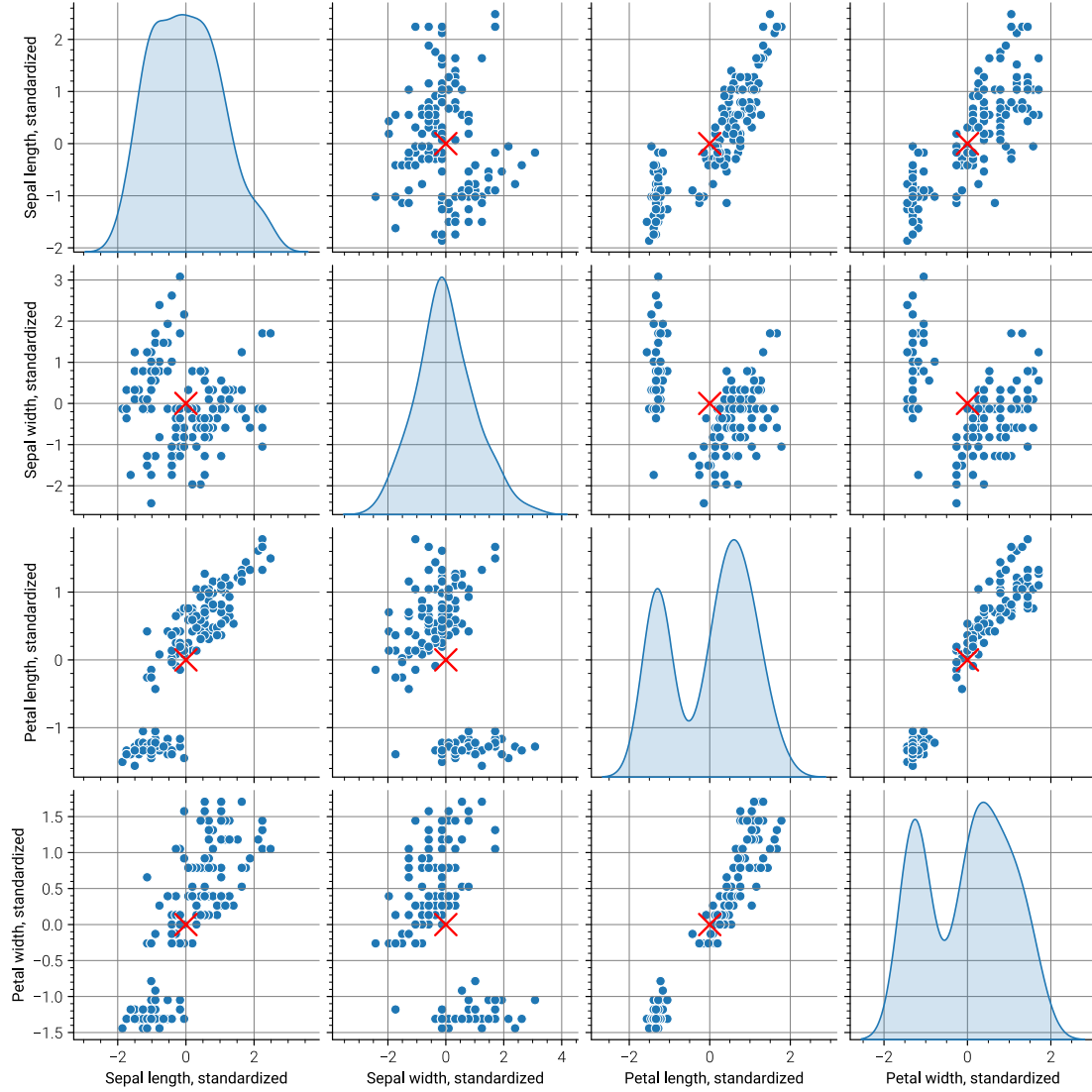


Figure 17. Pairwise Scatter Plot of the Iris Data (Standardized), with the Red $\times$ Marking the Centroid, Unitless

Linear Correlation Coefficient Matrix

The linear correlation coefficient matrix $\mathbf{P}$'s definition is:

$$\mathbf{P} = \begin{bmatrix} 1 & \rho_{1,2} & \cdots & \rho_{1,D} \\ \rho_{1,2} & 1 & \cdots & \rho_{2,D} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{1,D} & \rho_{2,D} & \cdots & 1 \end{bmatrix}$$

Figure 18 shows the iris data's correlation coefficient matrix $\mathbf{P}$. $\mathbf{P}$'s diagonal elements are all 1, and the off-diagonal elements are the pairwise correlation coefficients ρ_{ij} .

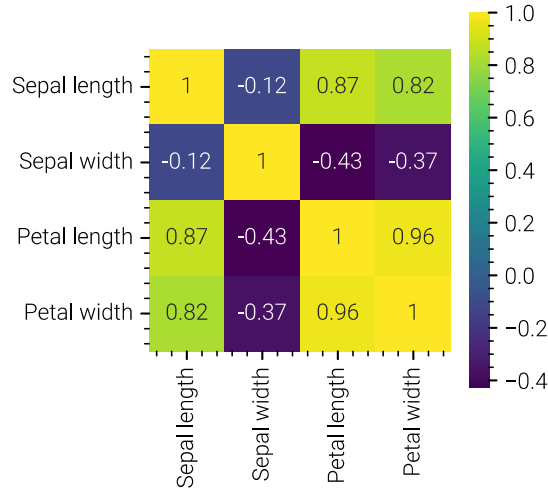


Figure 18. Linear Correlation Coefficient Matrix

The relationship between the covariance matrix Σ and the correlation coefficient matrix P is as follows:

$$\Sigma = \underbrace{DPD}_{D} = \underbrace{\begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_D \end{bmatrix}}_D \underbrace{\begin{bmatrix} 1 & \rho_{1,2} & \cdots & \rho_{1,D} \\ \rho_{1,2} & 1 & \cdots & \rho_{2,D} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{1,D} & \rho_{2,D} & \cdots & 1 \end{bmatrix}}_{\text{Correlation matrix, } P} \underbrace{\begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_D \end{bmatrix}}_D$$

Geometrically, the diagonal square matrix D in the equation above plays the role of scaling.

Figure 19 shows the heatmap of the relationship between the covariance matrix and the correlation matrix.

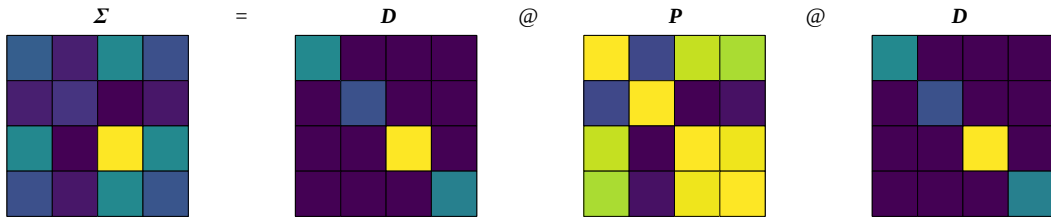


Figure 19. Heatmap of the Relationship Between the Covariance Matrix and the Correlation Matrix

From Σ , Deriving the Correlation Coefficient Matrix P

$$P = D^{-1} \Sigma D^{-1}$$

We're sure careful readers may have already noticed that the linear correlation coefficient matrix P is equivalent to the covariance matrix of the standardized data matrix, that is

$$P = \Sigma_z = \frac{\text{Gram matrix } Z^T Z}{n-1} = \frac{1}{n-1} \begin{bmatrix} \mathbf{z}_1^T \mathbf{z}_1 & \mathbf{z}_1^T \mathbf{z}_2 & \cdots & \mathbf{z}_1^T \mathbf{z}_D \\ \mathbf{z}_2^T \mathbf{z}_1 & \mathbf{z}_2^T \mathbf{z}_2 & \cdots & \mathbf{z}_2^T \mathbf{z}_D \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{z}_D^T \mathbf{z}_1 & \mathbf{z}_D^T \mathbf{z}_2 & \cdots & \mathbf{z}_D^T \mathbf{z}_D \end{bmatrix}$$

As shown in Figure 20, the equation above is also equivalent to computing a Gram matrix.

? Please analyze the operations corresponding to Figure 20's main-diagonal and off-diagonal elements, and their statistical meaning.

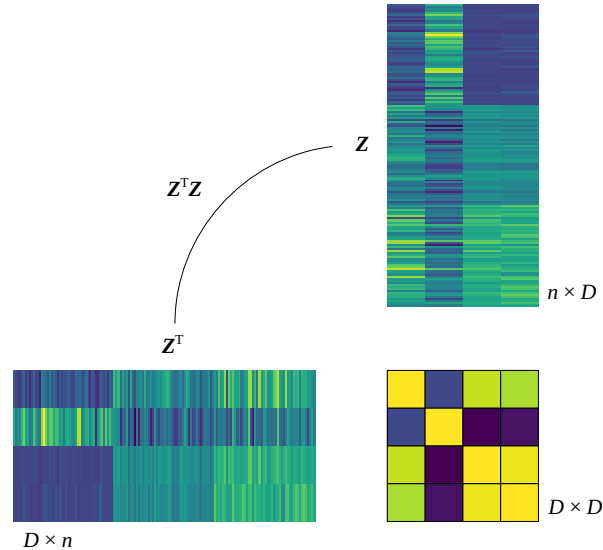


Figure 20. Computing the Standardized Data $\mathbf{Z}$'s Covariance Matrix ($\mathbf{X}$'s Linear Correlation Coefficient Matrix), Not Accounting for $n - 1$



Please study LA_12_01_01.ipynb on your own — it completes all the computations and visualizations in this section.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please modify LA_12_01_01.ipynb to perform an eigenvalue decomposition of the covariance matrix, and analyze the various properties of the eigenvalues and the eigenvector matrix.

Q2. Please modify LA_12_01_01.ipynb to perform an eigenvalue decomposition of the linear correlation coefficient matrix, and analyze the various properties of the eigenvalues and the eigenvector matrix.

Q3. Please study the broadcasting rule on your own:

<https://numpy.org/doc/stable/user/basics.broadcasting.html>